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# NMAP : THE BASICS :

Nmap is a open source network scanner that was first published in 1997.

It helps you to mainly do two things:

1 = Discovering live hosts on your networks or other network

2 = Discovering running services on a specific host.

- Nmap uses multiple ways to scan its targets:

IP range using "-" : To scan all the IP addresses from 192.168.0.1 to 192.168.0.10 write 192.168.0.1-10

IP Subnet using "/" : To scan a subnet write 192.168.0.1/24 equi: to 192.168.0.0-255

Hostname : You can specify the target by hostname

- Command used to scan network using nmap?

nmap -sn ipaddress range

when scanning a directly connected network where your device is directly connected to ethernet or wifi, Nmap starts by sending ARP requests, when a device responds to the ARP request, Nmap labels it as host is up.

- How to list that will be scanned if I run -sn command without actually scanning them?

nmap -sL ipaddr : Lists all the targets that will be scanned if i scan the network.

- How to scan the TCP ports of devices on a target network to check which ports are open or listening services?

nmap -sT ipaddr : will scan top 1000 tcp ports for devices
nmap -sT -p- ipaddr : will Scan all 65535 ports for devices
nmap -sT -p 1-1024 ipaddr : will scan ports from specified range
nmap -sT -F ipadd : will scan top 100 most common ports.

For checking for open ports it tries to complete the TCP three way handshake with every port.
If handshake successful: port open otherwise close
If TCP handshake is done and the port is open, it means nmap connected to that port, so it torn the connection established with open ports

- How to scan the UDP ports of devices on a target network to check which ports are open?
nmap -sU ipaddr

- SYN scan (Stealthy Scan): It is also used to Scan for open TCP ports on devices of a target network but unlike normal scan done using -sT, it doesn't make a three way handshake. It only sends a SYN packet if the target port replied it means it is listening and nmap end the connection without completing three way handshake.

Command: nmap -sS ipaddr

- How to detect Operating System for live hosts?
Command: nmap -sn -O ipaddr.

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- How to ~~???~~ tell the name of the service and service version listening on an open port?

Command: nmap -sT -sV ipaddr

-sT → scans open ports

-sV → tells the service name and version on that port

- Command : nmap -A 10.10.1.184

It stands for aggressive scanning. It does:
1 = port scanning for default tcp ports
2 = OS detection
3 = service & Version detection
4 = traceroute
this command do all these things in a single command

- Forcing the scan: If the target host doesn't reply during the host discovery phase i.e. if it doesn't reply to the ICMP packets nmap will mark it as down. But if I want to mark every host as online and port scan every host. (TCP port scans top 1000 ports)

Command: nmap -Pn

- How to control scan speed and timing?

slowest to Fastest

paranoid (0), Sneaky (1), polite (2), normal (3), aggressive (4), insane (5)

Command: nmap -sS -T0 or -T 0 or -T paranoid address

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- Probe = One test packet sent to : a port or a host or a service

--min-parallelism {numbprobes}
--max-parallelism {numprobes}

parallelism: Means the no. of packets or probes that are send at once not simultaneously to a service or a host or a port. Like example: If I set minimum as 50 and max as 100. So when I run the scan nmap sends greater than 50 and less than 100 probes at once and as soon as the packets get end they send new probes to keep the no. b/w 50 and 100

- --min-rate <number>
--max-rate <number>

It tells us how many new probes to send per second. You can set the maximum and minimum limit.

- --host-timeout <time> : This command is used to tell nmap the max time to spend on a particular host. If he doesn't reply mark him as timeout. This is good for slow hosts or host with slow network connection.

NOTE: nmap sends packets to host many times before marking it as down. But if the single packet replies the host is mark alive.

| | Host Discovery (By default) | Port Scan (By default) |
|---|---|---|
| nmap target | ✓ | ✓ |
| nmap -sn target | ✓ | ✗ |
| nmap -sT target | ✓ | ✓ |
| nmap -Pn target | ✗ | ✓ |

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- How can I get additional information or a real-time information while a scan takes place?

Command= nmap -v
↓
Verbosity

It shows real time information about what's happening in the scan.

-vv or -v2

-vvv or -v3 for more verbosity

- Saving scan report:

-oN filename: Normal output

-oX filename: XML output

-oG filename: grepable output (useful for grep or awk)

-oA filename: output all major forms

- Debugging: Enables debug output. Ex: It will show probes being sent, Replies received or missed, Timing decision many more

Command: nmap -d
↓
-d2, -d3 (for increasing debug verbosity)